

The impact of low ice cover on shoreline recession: A case study from Western Point Pelee, Canada

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ABSTRACT

The western shore of Point Pelee National Park (PPNP) in Ontario, Canada, exhibited high recession rates in the winter of 2005/2006. During the period 1959–2004, the western side of the park generally exhibited accretion, attributed to longer duration of ice coverage during winter and favorable topography and sedimentary conditions in comparison to the eastern side. Compared to the winter of 2003/2004, changes in Lake Erie water levels, winter wind regime and human-made structures and interferences were negligible during 2004–2006 within the Colchester-to-Southeast Shoal littoral cell in western Lake Erie, where PPNP is located. However, the winter of 2005/2006 exhibited overall low ice cover conditions unlike the typical freeze-up conditions in western Lake Erie. The 2006 high recession rates on the western shore of Point Pelee are attributed to the low ice cover conditions in the winter of 2005/2006. While less-than-average ice cover does not in its own cause erosion, in certain areas it increases the opportunity for sediment removal by waves by removing a significant source of protection during winter. The episode illustrates the potential impacts of ice-free and low ice cover winters on the western shore of Point Pelee and shores under similar conditions. Future climate change patterns are expected to result in shorter ice-durations and more frequent ice-free winters in the western basin of Lake Erie, exposing western PPNP to wave action during the winter season and resulting in increased erosion. The increased opportunity for shore erosion will negatively impact ecologically and economically viable spaces in PPNP.

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1. Introduction

Erosion occurs when shore sediments are lost at a rate faster than gained. Shorelines in many parts of the world have been retreating during the past several decades, including some shorelines that were previously prograding, (e.g. Deauville in France, Miami in Florida and Surfers Paradise in Australia; after Bird (2008)). At a given shore, only a few factors are responsible for the onset of erosion. In the Great Lakes, various factors may contribute to shoreline erosion such as storm and wind climate resulting in wave attack and sediment removal, rising lake levels resulting in submergence of the beach area and enhanced wave reach, and anthropogenic factors such as human-made coastal structures resulting in interruption of sediment supplies in the littoral cell.

During winter, ice cover may play a conflicting role in enhancing/inhibiting erosion. In periods with temperatures consistently below freezing, an ice foot forms along the beach as slush ice is driven against the shore by winds and waves (Hampton and Griggs, 2004), effectively blocking waves, freezing the beach area and preventing wave-driven loss of sediment. As ice melts and shifts, sediment

entrained in the ice is transported alongshore and offshore, resulting in local net sediment loss. There is therefore variability in the net effect of ice on shoreline erosion. In some cases, ice enhances erosion and in other cases it inhibits erosion. Barnes et al. (1994) found that nearshore ice plays a role in removing sediment from the southern Lake Michigan shore. Kempema et al. (2001) found that ice attached to the lake bed removes a significant amount of sediment from the nearshore zone of southwestern Lake Michigan. Other coastal studies on cold climate processes indicate that the net impact of thawing ice plays a role in removing sediment through sediment entrainment and transport offshore (e.g. Are et al., 2008; Bernatchez and Dubois, 2008). In Lake Erie along the Point Pelee shore, nearshore ice was found to protect the beach and where ice was absent, considerable erosion took place (after Coakley, 1977). Recently, Dale and Byrne (2010) found that during the subzero temperature periods of 2010 winter, water along the western beach of Point Pelee became frozen and was subsequently covered by snow, removing the possibility of any wave erosion during subzero temperature periods.

Through analyzing shoreline change rates and relevant conditions of the western shores of Point Pelee National Park (PPNP), Ontario, this study aims to assess the impact of low ice cover conditions on shorelines where winter ice forms regularly and potentially protects the shores from erosion due to winter storms. This study also aims to describe the conditions under which the net impact of winter ice

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cover will protect the shoreline rather than entrain and remove sediment. The study aims to increase the understanding of shoreline processes in western PPNP to assist managers in protecting and preserving this unique area.

2. The study area

Point Pelee National Park of Canada (PPNP hereafter), established in 1918, is the southernmost part of mainland Canada, extending 9.0 km from north to south and up to 4.2 km from east to west at its northern edge (Fig. 1). The park, a cusped foreland along the northern shoreline of Lake Erie, is subject to storm waves, storm surges, and seiches from all sides except its north. The PPNP is one of the most biodiverse national parks in Canada; about 70% of the park consists of marshes and the remainder is dryland forest, swamp forest, beach, and red cedar savanna (Trenhaile and Dumala, 1978; Smith and Bishop, 2002).

The park has been the subject of several studies regarding its origin, evolution, shoreline changes and related processes (Coakley, 1976, 1977; Trenhaile and Dumala, 1978; Coakley, 1980; Coakley et al., 1998; LaValle and Lakhan, 2000; Trenhaile et al., 2000; BaMasoud and Byrne, 2011). Recently, Hatfield et al. (2010) made magnetic measurements on sand samples from the eastern beach of PPNP to examine both the composition and spatial distribution of magnetic minerals. By mapping magnetic susceptibility along the eastern beach of PPNP, as a proxy for erosion, they identified 2.4 km (31%) of the beach where erosion was high.

Ecologically, PPNP is recognized as a Wetland of International Significance by the RAMSAR Convention of UNESCO (RAMSAR, 2010). However, the park has more species at risk than any other national park in Canada (Dobbie et al., 2006). Tourism in PPNP contributes to the regional economy as tens of thousands of birdwatchers are attracted to the park to watch the more than 380 species of migrating birds that fly through this location at the intersection of two major North American migratory flyways (Parks Canada, 2010). The study area, western PPNP, is of special economic significance because the majority of attractions for visitors to PPNP, including beaches, marsh access, hiking and bicycling trails, camps, picnic areas and the southern tip are along the western shore of the park (Parks Canada, 2010).

The study area, the western shore of PPNP (Fig. 1), is part of Colchester-to-Southeast Shoal littoral cell within the western basin of Lake Erie (Trenhaile et al., 2000). The general direction of longshore sediment transport is from west (Colchester Harbour) to east, due to the dominant westerly winds and long fetch from west to east (Reinders, 1988; Trenhaile et al., 2000). The bluffs between Colchester and Leamington (Fig. 1) consist of sand or sandy loam, providing a source for sediments transported eastwards. At the end of the littoral cell, the western shore of PPNP has historically been a depositional landform for most of the last 4000 years (Trenhaile et al., 2000).

The beach and offshore sediments consist mainly of eolian and shoreface sand. Having undertaken an extensive drilling program along the western barrier bar of PPNP, Coakley et al. (1998) showed non-cohesive sediments extending more than 300 m into the western basin of the lake (Figs. 9–11 in Coakley et al., 1998). During high-energy storms and highwater levels, sand migrates to the nearshore zone and gets deposited back during calm wind and low lake level periods.

The area contains several morphological features such as sand dunes, sand plains, sand ridges and swales (Fig. 2). Dunes in the north consist of eolian material on top of wave-deposited coarse sediments (Trenhaile et al., 2000). Small pebbles to coarse sand form ridges and swales to the south that are mainly wave-deposited, with fine eolian interbedded sediments. The sand plains, a low-relief area, consist of fine, well-sorted eolian sands, on top of coarser wave-deposited sands (Trenhaile et al., 2000).

Lake Erie at an average depth of 19 m is the shallowest of the Great Lakes. Extensive surface ice forms in the lake almost every winter (Environment Canada and U.S. Environmental Protection Agency, 1995). As the shallowest of the Great Lakes, Lake Erie is the most susceptible to storm surges (Danard et al., 2003). The western basin of Lake Erie, with an average depth of about 8 m and a maximum depth of 19 m, is shallower than the central and eastern basins. Therefore, during winter the western basin has the highest average ice cover concentrations among the three basins (Assel, 2004). A detailed bathymetry at for all of Lake Erie was completed by Holcombe et al. (2005) and is available at <http://www.ngdc.noaa.gov/mgg/greatlakes/erie.html>.

Within the study area, there are five small-craft harbors that act mainly as sediment sinks (Baird, 2008): Colchester Harbour, Cedar Beach Harbour, Kingsville Harbour, Leamington Harbour, and Sturgeon Creek Harbour.

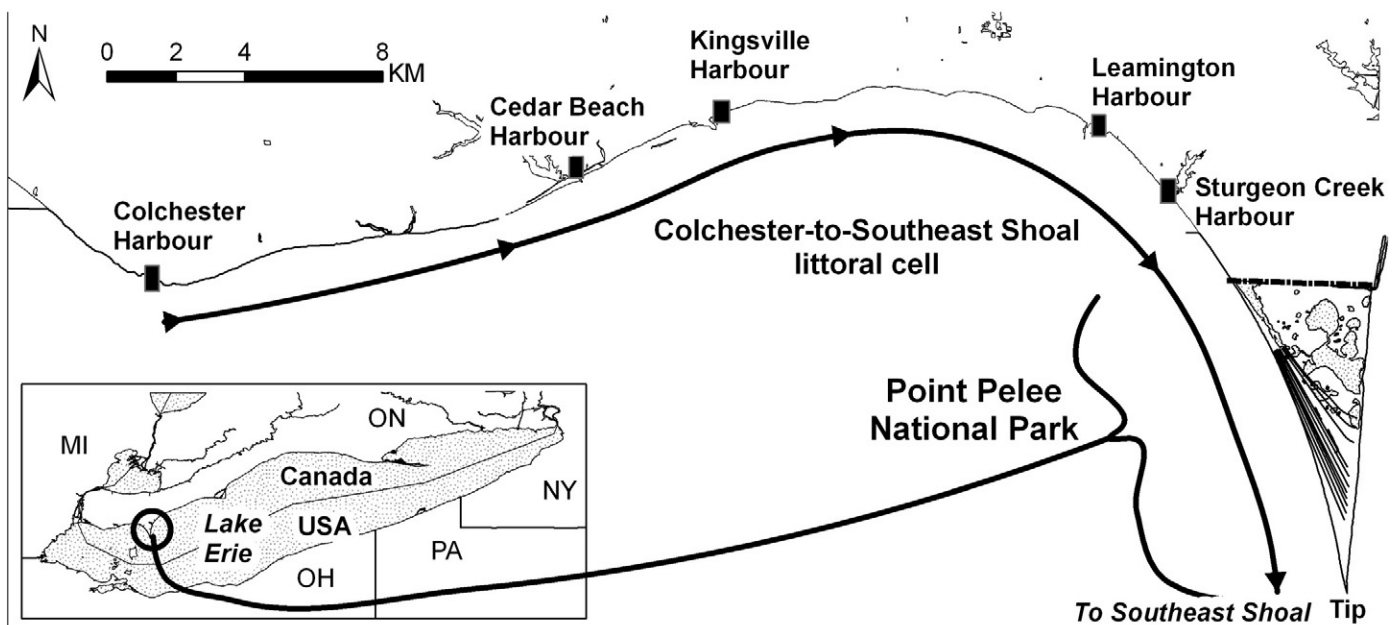


Fig. 1. Regional context of the study area with the littoral cell.

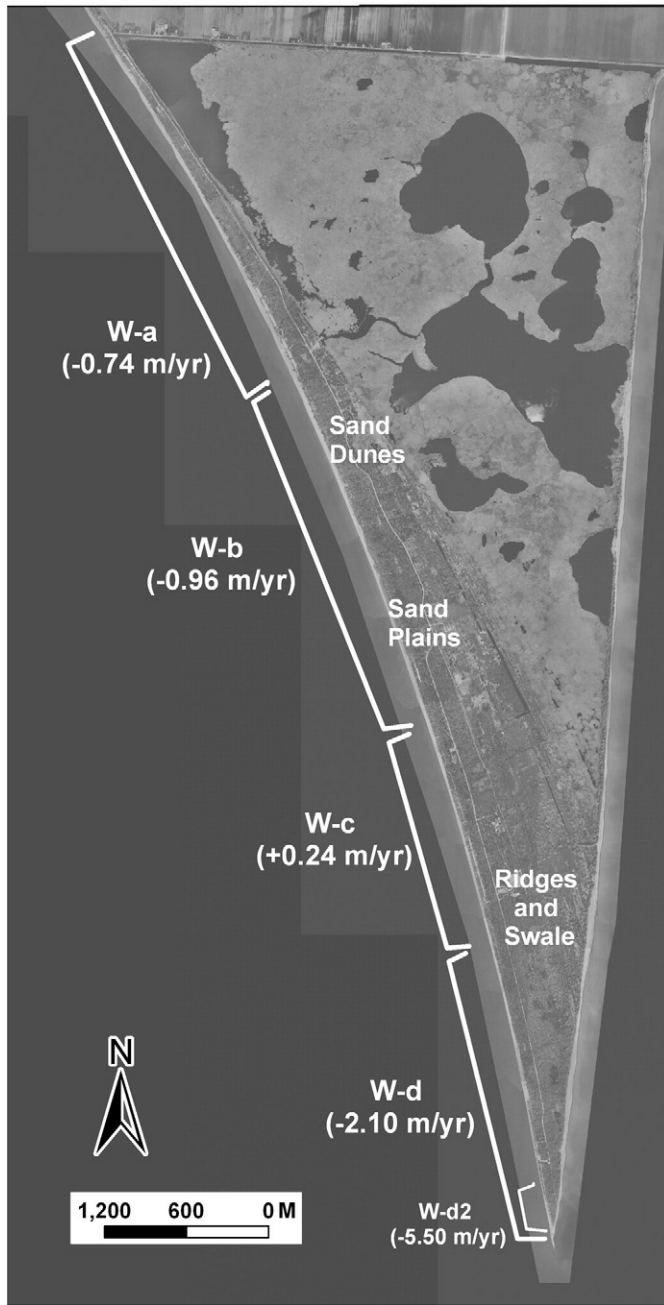


Fig. 2. End-point rates for Western Point Pelee shores during 2004–2006.

Beach Harbour, Kingsville Harbour, Leamington Harbour and Sturgeon Creek Harbour (Fig. 1). Sediment accumulation is evident on the western side of these harbors (Baird, 2008), confirming the dominant direction of transport in the littoral cell (e.g. Sturgeon Creek Harbour, Fig. 3). Despite these numerous sediment sinks, the western shore of PPNP has accreted for most of the 1959–2004 period, unlike PPNP's eastern shore which has been consistently eroding, resulting in a westward shift and net shrinkage of the park (Trenhaile et al., 2000; BaMasoud and Byrne, 2011).

3. Data and methods

Orthorectified aerial photos of Point Pelee from 2004 to 2006 were obtained from Parks Canada. The two sets of aerial photos were taken in April 2004 and April 2006, both at a scale of 1:10,000. Aerial photo resolutions were 0.50 m and 0.30 m, respectively. Shorelines were



Fig. 3. Sturgeon Creek Harbour, northwest of study area is typical in terms of pattern of sediment accumulation on its western side and of its size in comparison to the other four harbors in the littoral cell.

manually traced and digitized using Esri ArcGIS 9.2®. These shorelines were detected using the wet/dry boundary mark method (Boak and Turner, 2005). Following BaMasoud and Byrne (2011), 314 transects were created electronically every 30 m along the western shoreline perpendicular to a baseline parallel to the 1985 shoreline. Consequently, the 314 transects were divided into four sections based on the similarity in exhibited shoreline changes. To estimate shoreline change rates, measurements of the distance between digitized shorelines in 2004 and 2006 were taken along transects. Shoreline change rates [measured in m/y] were estimated using the end-point rate (EPR) method (as described by Dolan et al., 1991). Positive rates correspond to accretion and negative rates correspond to recession. End-point rates are then compared to rates from the period 1959–2004, analyzed in BaMasoud and Byrne (2011).

Publicly available hourly wind data for winters 2003/04 and 2005/06 were obtained from Canada's National Climate Data and Information Archive – Environment Canada, at www.climate.weatheroffice.gc.ca. Wind roses for the two seasons, November to May, were plotted and statistics of wind data were calculated for comparison purposes.

Monthly-averaged Lake Erie water levels were obtained from the NOAA Great Lakes Environmental Research Laboratory (GLERL) website, <http://www.glerl.noaa.gov>, collected by the United States National Ocean Service (NOAA, 2010). Annual lake level averages were calculated for the period 1959–2006. To contrast variability in average shoreline displacement and annual water level, both variables were standardized by removing temporal mean and dividing by standard deviation (similar to LaValle and Lakhani, 2000). Ninety-five percent of all dimensionless values should fall within ± 1.96 standard deviations of the mean.

Data on human-made structures and human-induced changes in the region were obtained from the local conservation authority, Essex Region Conservation Authority (ERCA) through personal communication. Empowered through provincial legislation, ERCA manages the natural resources of the Essex Region in partnership with member municipalities and the Province of Ontario (ERCA, 2011).

Ice-cover concentration data for four winters (2003–2006) were obtained from the United States National Ice Center website, <http://www.natice.noaa.gov>. Ice-cover charts used in this study consisted of 185 daily maps that were averaged into weekly fields. The digitized ice-cover charts for the Great Lakes are described by Assel (2003). Assel (2004) analyzed Lake Erie's ice cover for the period 1898–2002 using modeled data for winters 1898–1972 and observed data for winters 1973–2002. His analysis showed that the lake ice season for the western basin of Lake Erie usually starts in December and ends by April. Based on this definition of lake ice season, weekly-average ice cover concentrations were analyzed for four winters covering the period December 1st to April 30th of the years 2002 to 2006. There was no observed ice in May of any of these four winters.

Assel (2005) defined the seasonal average ice cover as the sum of the daily lake-averaged ice cover over a winter divided by 182 (the number of days between December 1st and May 31st). Analyzing ice cycles over a 30-winter base period (1973 to 2002) in the Great Lakes, he ranked ice cover cycles of Lake Erie into mild, typical and severe. These classifications are used in this study to describe the 2005/06 winter in Lake Erie. Median ice cover concentrations in the western basin of Lake Erie were averaged for the 2005/06 winter and compared to those of 2002/03, 2003/04 and 2004/05 winters.

4. Results and discussion

Shoreline change rates in Western Point Pelee are described for the period 2004–2006 in comparison to the 45-year averages from the period 1959–2004, which were previously analyzed by BaMasoud and Byrne (2011). The description is followed by a discussion of changes in factors that act on the shoreline of western PPNP shore in an attempt to explain the rates of recession during 2006 in relation to these changes. Comparison is made to conditions in the immediately preceding period (2000–2004) to contrast the conditions and changes of ice cover, Lake Erie water levels, wind regime and human interference in the two periods. Because storm-generated waves erode sections of the coastline or lakeshore in an immediate fashion, wind data are analyzed for two winters 2003/04 and 2005/06, for which post-winter aerial photo data are available. Annual averages of Lake Erie water level changes analyzed for 1959–2006 account for responses to interannual changes in lake levels. While engineered structures may impede sediment transport over the long term, the western PPNP shore exhibited negligible changes during the period of 2000–2004 and accreted during most of the period 1959–2000 (BaMasoud and Byrne, 2011). Therefore, investigation of human interference covers the period post-2004. Ice data are analyzed for the winter of 2005/06 in comparison to the three preceding winters. References are made to ice data analyzed previously by Assel (2004 and 2005) and Assel et al. (2003) covering the period 1898–2002.

4.1. Shoreline change rates in Western Point Pelee

Shoreline changes in the period 2004–2006 in western PPNP are different from those observed in the period 1959–2004. End-point rates for the period 2004–2006 show that the western PPNP shore exhibited recession (-0.97 m/y) along most of PPNP's shores (Fig. 2). In comparison, the western PPNP shore exhibited $+0.21$ m/y accretion over the 45-year period of 1959–2004 (Table 1). The only period during which the western PPNP shore exhibited documented recession in the 45-year period was 1977–1985, with an average rate of -0.87 m/y (BaMasoud and Byrne, 2011). The 1977–1985 period coincided with the high Lake Erie water levels, reaching levels unprecedented since 1959, which explained the high recession rate during this period.

Similar to the 1959–2004 period, shoreline change rates differed from one section of the study area to another during the 2004–2006 period. The northernmost section, W-a in Fig. 2, exhibited a net

recession rate of -0.74 m/y. Section W-b exhibited a net recession rate of -0.96 m/y. Section W-c exhibited a net accretion rate of $+0.24$ m/y. The most southernmost section (W-d), including the western side of the Point Pelee tip, exhibited the highest recession rate of -2.10 m/y, with the tip area receding at an average of -5.50 m/y. Changes from one section to another are attributed to short-term changes in littoral sediment transport. For comparison purposes, rates from the period 1959–2004 for these sections are provided in Table 1.

4.2. Physical processes and impacts on shoreline change rates

The following sections present a discussion of observed changes in ice cover, wind regime, Lake Erie water levels and human-induced changes during the winter of 2006.

4.2.1. Lake Erie wind-regime changes

Meteorological forcing in Lake Erie is responsible for wave action, storm surge and wave-generated currents. Mid-latitude cyclones, spanning from 1000 to 3000 km in diameter over the Great Lakes, generally move from west to east and provide the day-to-day variability in regional weather (Angel, 1996). The preferred path of cyclonic systems entering the PPNP area depends on the position of the polar front and the “upper westerlies,” both of which move to the south during winter (after Coakley, 1977). The orientation of Lake Erie is along the preferred path of prevailing cyclones, particularly impacting the eastern and western shorelines (after Angel, 1996). Therefore, the western shore of PPNP is directly exposed to waves resulting from the prevailing southwesterly and westerly (SW–W) winds during the winter season. Generally, these winds result in a southward sediment transport between Sturgeon Creek and the tip of PPNP (Coakley, 1977; see Fig. 1). Richards and Phillips (1970) analyzed overland winds near Lake Erie and synthesized wind characteristics over the lake for the years 1957 to 1966. Their analysis showed that during those ten years, SW to W winds on average represented 39.4% of all winds.

Intense storms in the Great Lakes occur from late fall throughout winter. Angel (1996) studied 100 storms with winds greater than 24.4 m/s in the Great Lakes basin over the period 1957 to 1985, and found that 92 of these storms were cyclones. This high frequency of cyclone occurrence among storms underscores the impact of SW–W winds and resulting waves on the western shore of PPNP. Angel (1996) found that 76 of the 92 cyclones occurred between November and March. This study compares wind data from November to May for the 2003/2004 and 2005/2006 seasons, with a focus on SW to W winds (Figs. 4, 5) to explore if the 2005/2006 season exhibited higher than usual wind speeds or more frequent storms.

Statistics for these two seasons are summarized in Table 2 and their windroses are shown in Figs. 4 and 5. Southwesterly to westerly winds were more frequent in 2005/06. In 2005/06 SW to W winds accounted for 35.1% of all winds, while in 2003/04, 30.5% came from that quadrant. However, the statistics show that in 2003/04 the high-speed SW and W winds (>8 m/s and >10 m/s), which are more likely to cause higher energy waves, all things being equal, were stronger and more frequent (Table 2). Events of SW to W winds that are of high speed (>8 and >10 m/s) were more frequent, with 262 in 2003/04 compared to 226 in 2005/06. Low ice cover in the western basin during the 2006 winter may have enhanced wind speed. According to Angel (1996), winter cyclones intensify under strong temperature contrasts between cold air and the ice-free water surface. Despite this potential intensification, the weighted average speed of winds from SW to W was higher in 2003/04 than in 2005/06, negating some of the impact resulting from more frequent SW–W winds in 2005/06. The weighted average speed of SW to W winds was 5.51 m/s in 2003/04 compared to 5.15 m/s in 2005/06. In both seasons, SW to W winds were less frequent than those derived

Table 1

Comparison of average shoreline change rates in Western PPNP shores by sections (refer to Fig. 3 for locations). Average end point rates for 45 years are provided for comparison. Sections W-d1 and W-d2 constitute section W-d.

Section on PPW	Average EPRs 1959–2004 (BaMasoud and Byrne, 2011)	Average EPRs 2004–2006
W-a (transects 1–89)	+0.20	−0.74
W-b (transects 90–182)	+0.40	−0.96
W-c (transects 183–237)	+0.23	+0.24
W-d (transects 238–314)	−0.01	−2.10
W-d1 (transects 238–305)	−0.03	−1.65
W-d2 (transects 306–314, the Tip)	+0.14	−5.50
Average across PPW	+0.21	−0.97

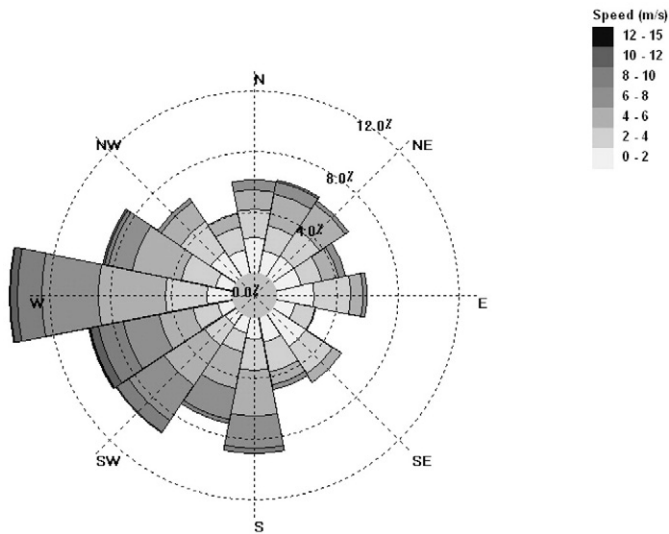


Fig. 4. Wind rose for Winter 2003/04, derived from hourly wind data available from Canada's National Climate Data and Information Archive.

by Richards and Phillips (1970). Despite being subjected to SW to W winds with higher weighted average speed than in 2005/2006, the western PPNP shore in 2004 exhibited negligible change (BaMasoud and Byrne, 2011).

4.2.2. Lake Erie water level changes

Changes in Lake Erie water levels are among the principal variables that influence shoreline changes in Point Pelee (Coakley, 1976; Trenhaile and Dumala, 1978; LaValle and Lakhan, 2000). Generally, the shoreline change rate of western PPNP exhibits inverse correlation to lake level (BaMasoud and Byrne, 2011). In periods of rising lake levels (e.g. 1977 to 1985 and 2000 to 2004), the western PPNP shoreline receded, and vice versa during periods of declining lake levels (e.g. 1985–2000, Fig. 6). In April 1985, Lake Erie water levels reached a high of 174.98 m above chart datum, referenced to International Great Lakes Datum 1985 (IGLD 1985). As a result, the western PPNP shore experienced recession in all of its sections in that year. In comparison, the western PPNP shore experienced higher recession rates in 2006 than in any of the aforementioned periods of rising lake levels despite the minor increase of 0.03 m in the annual

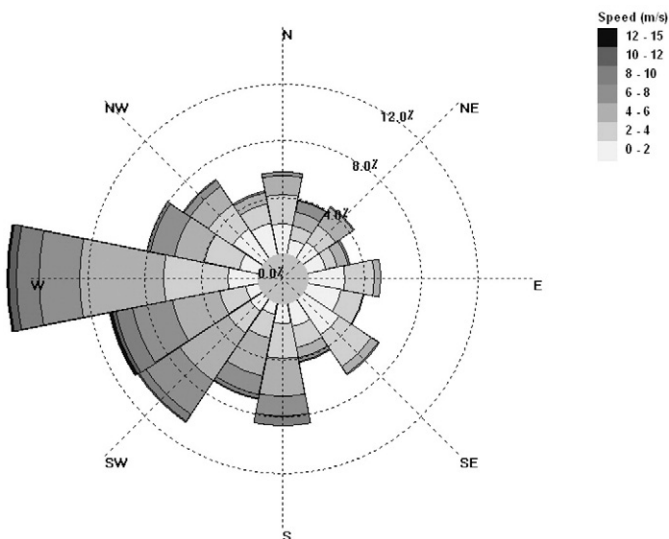


Fig. 5. Wind rose for winter 2005/06, derived from hourly wind data available from Canada's National Climate Data and Information Archive.

Table 2

Statistics of wind data for winter 2003/04 and 2005/06.

Year (Nov to May)	2003/04	2005/06
Total number of hourly measurements	4368	4344
Data recovery	98.86%	99.13%
Percentage of winds from SW–W	30.50%	35.09%
Weighted average speed of winds from SW–W	5.51 m/s	5.15 m/s
Events of SW–W winds > 8 m/s	213	179
Events of SW–W winds > 10 m/s	49	47

average lake level in 2006 relative to 2004 (Fig. 7). A linear regression model based on the standardized historical lake levels and observed shoreline changes from 1959 to 2004, estimates that the lake level would have to increase by 0.31 m to a mean of 174.46 m to cause the observed recession of 2006. The model also estimates that based on the observed increase in lake level in 2006, the annual average recession rate would have been 0.02 m/y, a very small rate that is within error estimates.

In summary, despite the strong relationship between rising lake level and shoreline recession observed in previous studies (Trenhaile et al., 2000; BaMasoud and Byrne, 2011), a 0.03 m increase in lake level is too small to account for the change on its own.

4.2.3. Human-induced changes

Human-made structures intended for shoreline protection can cause immediate interruption of sediment supply at the downdrift end of the littoral cell. For example, shoreline protection structures were installed in 1973 north of the eastern side of PPNP. The installations resulted in obstructing sediment flow toward PPNP's eastern shores and caused up to 200 m of recession within the following four years along the northern shores of PPNP's eastern side (Coakley, 1977; LaValle and Lakhan, 1997; BaMasoud and Byrne, 2011). In the Colchester-to-Southeast Shoal littoral cell, there were no significant new human-made structures that may have significantly disrupted sediment movement in the littoral cell. Until 2005, Kingsville Harbour was not a regulated harbor. In 2005 the local conservation authority, ERCA, required that sand dredged from Kingsville Harbour be deposited on the downdrift side to minimize the effects of littoral cell interruption. The effect of such regulation would likely be an increase in the supply of sediment downdrift. However, there are at least two sink areas downdrift in the littoral cell resulting from the existence of two small harbors to the east of Kingsville Harbour, reducing the impact of the increased supply of sediments. According to ERCA, no other new structures or interferences occurred during the period 2004–2006 (Tim Byrne, Personal Communication, 2010).

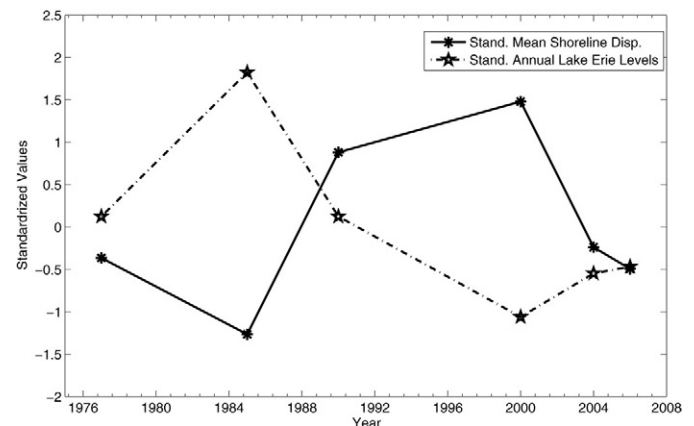


Fig. 6. Relationship between standardized values for mean shoreline displacement and annual Lake Erie water levels.

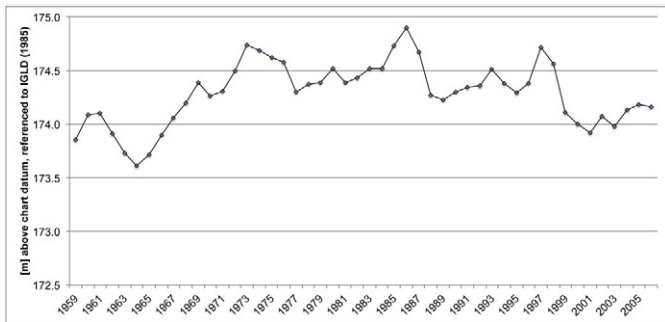


Fig. 7. Lake Erie water levels 1959–2008. Elevation in meters above chart datum, referenced to IGLD 1985.

Since there were no significant changes in the *status quo* of human structures and no new interferences within the littoral cell that reduce sediment supply to the western shore, the 2006 recession is probably not attributable to human-induced changes within the Colchester-to-Southeast Shoal littoral cell.

4.2.4. Lake Erie ice cover

The severity of lake ice season can be described by the seasonal average ice cover. The annual maximum ice concentration (AMIC) is the greatest percentage of lake surface area covered by ice each winter. Based on Assel's (2005) classifications, the ice cycle in Lake Erie during winter 2005/06 was mild at 4.8% ice cover seasonal average, the fourth lowest of the previous 34 winters (1973–2006), with most ice concentrated in the western basin (Figs. 8, 9). Although the 2006 was not a typical winter in terms of ice cover, the low ice cover was not unprecedented. The annual maximum ice concentration (AMIC) is the greatest percentage of lake surface area covered by ice each winter. The AMIC for Lake Erie in winters 1983, 1991 and 1998, were well below the 94% median AMIC for 1963–2001 in Lake Erie indicating relatively low ice coverage during these three winters (Assel et al., 2003). The unavailability of aerial photos for the years 1983, 1991 and 1998 ruled out the possibility of isolating the impacts of the ice cover conditions on shoreline changes in these three years.

During mild winters, Lake Erie's western basin typically would have higher ice coverage than the central or eastern basins because of its relative shallowness. Median ice cover concentrations in the western basin of Lake Erie were averaged for the winters 2002/03, 2003/04 and 2004/05 along with median ice cover concentrations of 2005/06 winter (Fig. 10). In the three winters 1983, 1991 and 1998,

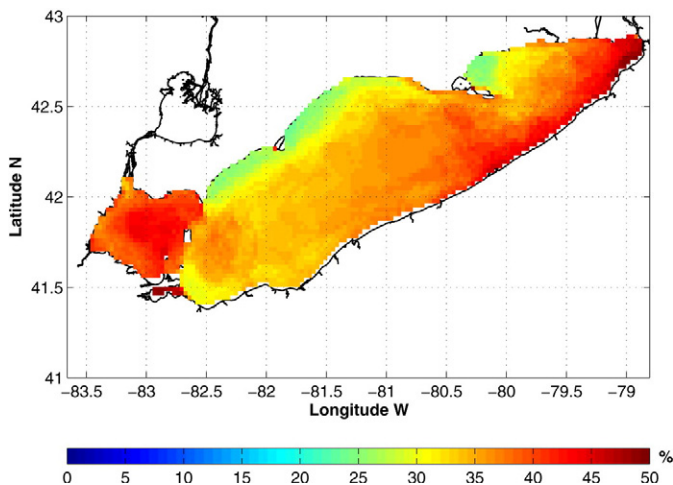


Fig. 8. Average ice cover concentration for winters of 2002/03, 2003/04 and 2004/05, obtained from the United States National Ice Center.

lake surface freezing started around mid-December and remained frozen through March, consistent with ice cover patterns for the years 1973 to 2002 (Fig. 7a in Assel, 2004). In comparison, during the winter of 2005/06, the western basin exhibited a brief period of extensive ice cover in late December. During most of January, and all of February and March 2006, the western basin was either ice-free or had little ice coverage (0–30%) (Fig. 10).

Unlike the three winters preceding it, the winter of 2005/06 exhibited relatively low ice cover so western PPNP shore was exposed to storm-generated waves most of winter. These anomalous ice conditions may explain the high rates of erosion during the 2005/06 winter. As a result of the difference in ice cover regime during the two years, the western PPNP shores experienced different amounts of net storm energy over the two years during ice-free periods. Over 90% ice cover concentrations during January and February 2004 meant that wind-generated waves were virtually blocked during the storm season. As described above, during most of the season, ice cover during winter 2006 was non-existent and waves had direct access to shores during all of the winter season, exacerbating erosion along the western PPNP shore.

Due to the shallowness of the western basin of Lake Erie, ice conditions are more sensitive to changes in atmospheric forcing and air temperature, resulting in quick freezing in the beginning of winter and quick melting at its end (Assel, 2005), and thus minimizing the impact of sediment removal by ice blocks that drift into deeper water. Further, the shallowness of the basin ensures that sediment-laden ice does not drift too far from the shoreface before melting and depositing the sediments in areas where sediments can be brought back to the beach during the summer fair weather period.

Ice-free conditions in Lake Erie winters are similar to those predicted under global greenhouse warming scenarios (Assel, 1991). Under the global greenhouse warming scenario modeled by the Canadian Centre for Climate Modeling and Analysis (CGCM1), severe reductions in ice cover will result from temperature increases and the frequency of ice-free winters is projected to increase such that 74% of winters in the western basin of Lake Erie will be ice-free by 2090 compared to an average of 2% for the period 1951–1995 (Lofgren et al., 2002). The model also projects a decrease in average ice duration from 91 days for the period 1951–1995 to 27 days by 2090 under warming conditions. Similar trends have been observed in the Arctic Sea ice extents for the period 1979–2006. Cavalierie and Parkinson (2008) analyzed 28 years of Arctic sea ice extents and areas derived from satellite passive-microwave radiometers and found a decreasing trend in ice extents for the Northern Hemisphere overall. In the Great Lakes region, results of an analysis of recent (1951–2000) trends in freeze-up and

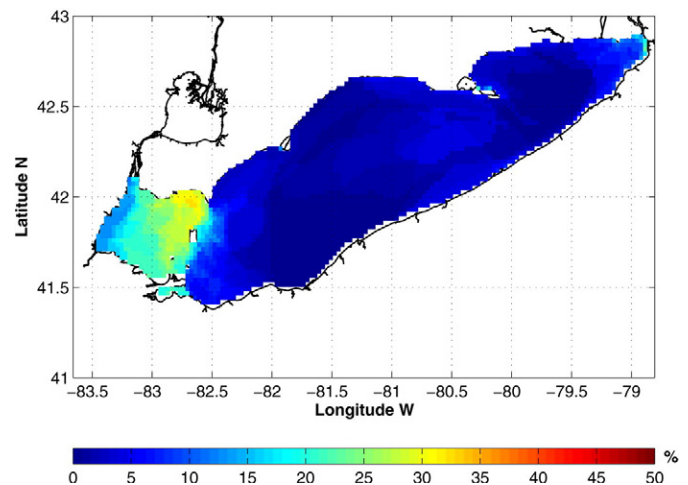


Fig. 9. Average ice cover concentrations in winter 2005/06, obtained from the United States National Ice Center.

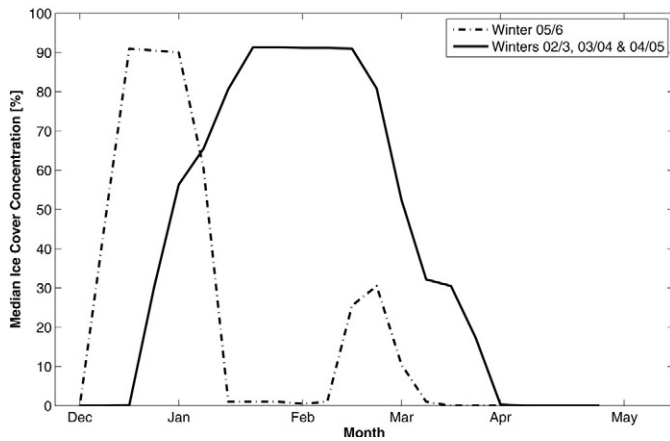


Fig. 10. Median ice cover concentrations (%) in winter 2005/06 versus winters 2002/03, 2003/04 and 2004/05 in the western basin of Lake Erie.

break-up dates showed trends toward later freeze-up and earlier break-up dates, leading to shorter ice duration (Duguay et al., 2006). These trends will probably result in net recession for the western PPNP shorelines, reversing the accretion trend observed during the period 1959–2004, especially given the existence of five harbors that reduced the amount of sediments available for deposition on the western PPNP shores. While less-than-average ice cover does not in its own cause erosion, in certain areas it increases the opportunity for sediment removal by waves by removing a significant source of protection during winter. As of April 2008, the shores have not recovered of this erosion episode.

As suggested from western PPNP physical conditions discussed above, a shore that exhibits the conditions below would probably experience erosion inhibition as a net impact of winter ice-coverage.

1. The shore is shielded from waves for extended periods during winter by shore ice coverage. In western PPNP (December through April, 1973–2002), shores were on average protected for 62 days each winter with lake ice that was at least 90% concentration of lake surface area (Assel et al., 2003; BaMasoud and Byrne, 2011).
2. The shore experiences winters accompanied by highest storm intensity and frequency compared to the other seasons. In western PPNP, most of the storms occur starting from late fall and continue throughout winter. During other seasons, opportunities for sediment deposition at the shore occur as lower energy waves that bring sediments forward to the shore.
3. The shore is directly facing winter storm and high speed winds. The western PPNP shore is directly exposed to waves resulting from its orientation that faces the prevailing SW–W winds during the winter season. These winds result in direct wave attack and southward sediment transport that removes sediments southward, unless blocked by winter ice.
4. The shore consists mainly of unconsolidated sediments that can migrate offshore and onshore during active and calm wave conditions. Fine sediments such as clay and silt are lost permanently from a littoral cell once entrained by waves (e.g. eastern PPNP shore), while sand gets deposited nearshore during summer and low lake levels as on the western PPNP shore.

5. Conclusions

The 2006 recession episode illustrates the impacts of a winter with low ice cover on the western shores of Point Pelee. The western PPNP shore is historically an accreting shoreline for most of the last 4000 years. However, during the winter of 2006, most of the western PPNP shoreline exhibited heightened recession rates (an average of -0.97 m/y), not observed since 1959. The minimal ice cover during

the 2006 winter exposed PPNP's shores to wave action throughout winter, probably resulting in the observed shoreline retreat. In comparison, the 2004 winter was typical with freezing conditions across the western basin of Lake Erie throughout winter, leading to consistent ice cover blocking most of SW–W wind-generated waves. Barnes et al. (1994) have shown that ice rafting of sand during spring break-up can deliver significant amounts of sediment far offshore and hence contribute to shoreline erosion. However, this effect should be relatively minor in cases like Point Pelee where ice both forms and melts rapidly and ice rafting tends to deliver sediment to the near-shore zone.

During the 2004 to 2006 period, there were no significant changes in the wind regime, Lake Erie water levels or human-made structures or interferences. The frequency and speed of SW–W winds were comparable in the two winters of 2003/04 and 2005/06. Annual average of lake water level changed by only 0.03 m, too little to trigger the 2006 recession event. No new human-made structures were installed in the period 2004–2006.

Recent observations of the Great Lakes' ice duration and climate change projections indicate a trend of shorter ice durations and more frequent ice-free winters in Lake Erie. If these trends continue, the western shores of Point Pelee and areas under similar conditions in the Great Lakes will be exposed to winter storm wave action throughout the season. Under such conditions, the previously accreting western PPNP shore will probably continue to recede following the 2006 recession event, resulting in wide areas of the park's ecologically and economically viable spaces being lost to the lake as a result of enhanced erosion. The implications for the impact of low ice cover on coasts with similar conditions are increased opportunities for shoreline erosion.

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